

EMILIO JOSEPH CALMA

Software Engineer · Backend & Applied Machine Learning

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PROFESSIONAL SUMMARY

Software Engineer and Deakin University student with 3+ years of self-directed development experience. Focused on backend systems, applied machine learning, and AI-integrated applications — independently studying ML and AI beyond university curriculum. Published on the Microsoft Store. Looking for a junior software engineering role where I can work on hard problems and grow fast.

FEATURED PROJECT

Sentiment Analysis API *FastAPI · PyTorch · HuggingFace Transformers · Docker · Prometheus/Grafana*

sentiment-api-nv4e.onrender.com/docs

Repo: github.com/H41s3/sentiment-api · Live: sentiment-api-nv4e.onrender.com/docs

- ▶ Architected a layered production ML-serving API (FastAPI + quantized DistilBERT) with request-scoped async offloading so CPU-bound inference never blocks the event loop
- ▶ Debugged a real production memory crash under a hard 512MB deployment ceiling — isolated root cause, tested three fix candidates, verified final result at ~442MB stable against live platform metrics
- ▶ Designed strict Pydantic API contracts, opt-in API-key auth, and per-key rate limiting with a Redis-backed distributed counter to fix a multi-worker rate-limit drift bug
- ▶ Shipped full observability (custom Prometheus metrics, provisioned Grafana dashboards) and a 380+-test suite enforced by a 95% coverage gate in CI

OTHER PROJECTS

p1p — Desktop Companion App *Electron · JavaScript · HTML/CSS · GitHub Actions · electron-builder*

[Microsoft Store](#)

- ▶ Built a BB-8-style system tray companion app that sends gentle wellness nudges throughout the workday — stretch reminders, hydration prompts, motivational boosts
- ▶ Packaged for Windows and Mac using electron-builder; automated Microsoft Store builds via GitHub Actions CI/CD pipeline
- ▶ Originally built as a personal gift; polished and published publicly on the Microsoft Store after successful real-world testing

[In development](#)

wtf — Screen-Time iOS App *Swift · SwiftUI · FamilyControls · DeviceActivity*

- ▶ Building an on-device iOS app that fights doomscrolling: a mascot cat reacts to screen time on chosen apps and shields them past a limit, using Apple's Screen Time frameworks (no backend, no data leaves the phone)

cu1sinefy.netlify.app

Cuisinefy — Recipe Discovery App *TypeScript · React · Edamam API*

- ▶ Built a recipe search interface consuming the Edamam API with multi-filter support across diet type, cuisine, and macros

p3tpals.netlify.app

PetPals — Client Pet Shop Website *JavaScript · React · React Router · Vite*

- ▶ Built a multi-page front-end UI for a client's pet shop (pet listings, product pages, cart, blog, contact) with client-side routing; deployed on Netlify

TECHNICAL SKILLS

Languages	Python, JavaScript / TypeScript, SQL, Swift
Backend	FastAPI, Node.js, SQL, RESTful API design, Pydantic, Redis, Docker
AI / ML	PyTorch, HuggingFace Transformers, TensorFlow, scikit-learn, NumPy, Pandas, OpenAI API
Mobile	iOS development (Xcode), Electron (Windows/Mac desktop)
DevOps	Git & GitHub, CI/CD, GitHub Actions, Prometheus/Grafana, Render, Netlify
Frontend	React, Vite, TailwindCSS, Shadcn UI

EDUCATION

Deakin University — Bachelor of Software Engineering (Honours)

2025 – 2028

Deakin College — Diploma of Information Technology

2024 – 2025

WORK HISTORY

The Butchers Club — Retail Sales & Hygiene Assistant

Apr 2024 – Jun 2026

- ▶ Delivered customer service, managed POS transactions, and maintained food-safety compliance under peak-hour pressure

TEN BBQ & Hotpot — Kitchen Staff

Dec 2023 – Apr 2024

- ▶ Operated in high-volume service, coordinating with front-of-house staff and maintaining prep standards under continuous load

ADDITIONAL

Available for full-time, part-time, or internship roles.